

Introduction to Machine Learning

Week 3

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1. (1 Mark) For a two-class problem using discriminant functions (δ_k - discriminant function for class k), where is the separating hyperplane located?

- (a) Where $\delta_1 > \delta_2$
- (b) Where $\delta_1 < \delta_2$
- (c) Where $\delta_1 = \delta_2$
- (d) Where $\delta_1 + \delta_2 = 1$

Soln. C

2. (1 Mark) Given the following dataset consisting of two classes, A and B , calculate the prior probability of each class.

Feature 1	Class
2.3	A
1.8	A
3.2	A
2.7	B
3.0	A
2.1	A
1.9	B
2.4	B

What are the prior probabilities of class A and class B ?

- (a) $P(A) = 0.5$, $P(B) = 0.5$
- (b) $P(A) = 0.625$, $P(B) = 0.375$
- (c) $P(A) = 0.375$, $P(B) = 0.625$
- (d) $P(A) = 0.6$, $P(B) = 0.4$

Soln. B

3. (1 Mark) In a 3-class classification problem using linear regression, the output vectors for three data points are $[0.8, 0.3, -0.1]$, $[0.2, 0.6, 0.2]$, and $[-0.1, 0.4, 0.7]$. To which classes would these points be assigned?

- (a) 1, 2, 1
- (b) 1, 2, 2
- (c) 1, 3, 2
- (d) 1, 2, 3

Soln. D

4. (1 Mark) If you have a 5-class classification problem and want to avoid masking using polynomial regression, what is the minimum degree of the polynomial you should use?
- (a) 3
 - (b) 4
 - (c) 5
 - (d) 6

Soln. B

5. (1 Mark) Consider a logistic regression model where the predicted probability for a given data point is 0.4. If the actual label for this data point is 1, what is the contribution of this data point to the log-likelihood?
- (a) -1.3219
 - (b) -0.9163
 - (c) +1.3219
 - (d) +0.9163

Soln. B

6. (1 Mark) What additional assumption does LDA make about the covariance matrix in comparison to the basic assumption of Gaussian class conditional density?
- (a) The covariance matrix is diagonal
 - (b) The covariance matrix is identity
 - (c) The covariance matrix is the same for all classes
 - (d) The covariance matrix is different for each class

Soln. C

7. (1 Mark) What is the shape of the decision boundary in LDA?
- (a) Quadratic
 - (b) Linear
 - (c) Circular
 - (d) Can not be determined

Soln. B

8. (1 Mark) For two classes C_1 and C_2 with within-class variances $\sigma_{w1}^2 = 1$ and $\sigma_{w2}^2 = 4$ respectively, if the projected means are $\mu'_1 = 1$ and $\mu'_2 = 3$, what is the Fisher criterion $J(w)$?
- (a) 0.5
 - (b) 0.8
 - (c) 1.25

(d) 1.5

Soln. B

$$S_w = \sigma_{w1}^2 + \sigma_{w2}^2 = 1 + 4 = 5 \quad S_b = (\mu'_2 - \mu'_1)^2 = (3 - 1)^2 = 4 \quad J(w) = \frac{S_b}{S_w} = \frac{4}{5} = 0.8$$

9. (2 Marks) Given two classes C_1 and C_2 with means $\mu_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\mu_2 = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ respectively, what is the direction vector w for LDA when the within-class covariance matrix S_w is the identity matrix I ?

(a) $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$

(c) $\begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix}$

(d) $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$

Soln. D

$$S_w \propto \mu_1 - \mu_2$$